

## Investment Recommendation

The results of this research indicate that historical returns are not the only criterion to use when making investment decisions; other factors such as risk, diversification and changing macro-economic conditions should also be taken into account. Bitcoin had the largest maximum return, the greatest volatility and the greatest downside risk over the sample period of the years 2020 to 2025 of the selected assets. BDO, on the other hand, had the highest risk-adjusted return, and had the lowest correlations with all of the other assets, making its greatest contribution to portfolio efficiency, not maximum absolute returns. The analysis concluded that Apple had provided significant long term capital gains with lower volatility as compared to Bitcoin, while SPY offered wide exposure to the U.S. equity market and was the benchmark for the whole analysis. The exploratory analyses and SQL investigations together show that each asset performs differently in different markets, underscoring the value of diversifying away from investing exclusively in a single high-performing asset.

Based on these findings, this study recommends constructing a diversified portfolio. Given these results, this research suggests the development of a diversified portfolio with a good mix of growth and effective risk management. The proposed allocation is based on a mix of analyses of total returns, risk-adjusted performance, cross-asset correlations, tail risk and macroeconomic sensitivity rather than historical returns.

| Asset                        | Proposed Allocation | Investment Amount |
|------------------------------|---------------------|-------------------|
| Apple (AAPL)                 | 35%                 | USD 350,000       |
| BDO Unibank                  | 30%                 | USD 300,000       |
| SPDR S&P 500 ETF Trust (SPY) | 20%                 | USD 200,000       |
| Bitcoin (BTC)                | 15%                 | USD 150,000       |

The proposed allocation is not a result of a quantitative portfolio optimization model, but is meant as a practical allocation suggestion. The allocation is highest for Apple, due to their strong long-term growth but shorter risk than Bitcoin. The high risk-adjusted return noted by BDO, and the fact that its correlations with the other assets were generally low, results in a large allocation for this asset. SPY offers diversified exposure to the U.S. equity market at the same time that it helps to minimize the concentration risk of investing in individual securities. Lastly, Bitcoin gets the least allocation to account for its outstandingly high returns, while at the same time its exposure to the volatility and tail risk of its portfolio is limited. In summary, the recommended allocation is consistent with the study's key finding: "A strategy for balancing return, risk, and diversification" is a more sustainable investment strategy than "The maximization of historical returns.

## **Conclusion**

This study was able to create a consolidated financial database consisting of four financial assets (Apple Inc (AAPL), BDO Unibank (BDO), Bitcoin (BTC) and the SPDR S&P 500 ETF Trust (SPY)) and four macroeconomic indicators from the Federal Reserve Economic Data (FRED) database. The study developed a consistent analytical framework over the 2020-2025 period by systematically acquiring, cleaning, standardizing the calendar, filling in the missing data, integrating the data into a database, and conducting an analysis using SQL.

The empirical results show that returns are not the only measure that can be used to assess the performance of investment. As is evident, Bitcoin had the largest risk (downside) along with the highest volatility and the highest cumulative return, which is a balance between the risk and the return. BDO, on the other hand, generated the highest risk adjusted return and consistently had the lowest correlation with the other assets, highlighting its ability to be a good diversification asset. However, Apple's long-term growth prospects were high and its risk levels were relatively low, whereas Bitcoin was considered as a risky asset. Moreover, Apple's long-term growth prospects were high while Bitcoin's were risky, and SPY provided diversified exposure to the broader U.S. equity market and served as a stable benchmark for asset performance evaluation. Together, this suggests that no single type of asset dominates in every respect and that a diversified portfolio of assets is likely to be the best approach to achieving investment goals.

The analyses also demonstrated that asset performance was different under various macroeconomic conditions, such as high market volatility, inflation, and interest-rate fluctuations. The exploratory analyses and SQL investigations repeatedly show that the performance of the assets is influenced differently in the different macroeconomic conditions, thus showing that one should not only consider the past performance of the assets, but their performance in the wider economic context.

Similarly, the modest associations found among BDO and the rest of the assets suggest that diversification still has a role to play in decreasing the overall risk of a portfolio.

Overall, the results confirm the tenets of Modern Portfolio Theory and show that the key to a good investment portfolio is not necessarily to maximize the performance of any individual asset but to balance the expected return, risk and diversification among the assets. Bitcoin had the highest level of capital appreciation, but its high volatility makes it poor as a primary holding in a portfolio. However, the comparatively small cumulative return in the BDO asset class does not diminish the value of the asset class since its superior risk adjusted performance and diversification benefits increase the overall efficiency of the portfolio. Hence, a multi-asset approach with exposure to Apple, BDO, SPY, and a measured exposure to Bitcoin offers a more balanced approach to long-term investing, as compared to a capital focus in a single asset class. Although the prediction of market outcomes from historical performance is not certain, the analytical framework that has been developed in this study shows how integrated financial databases, and SQL based financial analysis can contribute to more informed and evidence based investment decision making.